



Hierarchical Bayes estimation of spatial statistics for rates

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ABSTRACT

The U.S. Bureau of Labour Statistics publishes monthly unemployment rate estimates for its 50 states, the District of Columbia, and all counties, under Current Population Survey. However, the unemployment rate estimates for some states are unreliable due to low sample sizes in these states. Datta et al. (1999) proposed a hierarchical Bayes (HB) method using a time series generalization of a widely used cross-sectional model in small-area estimation. However, the geographical variation is also likely to be important. To have an efficient model, a comprehensive mixed normal model that accounts for the spatial and temporal effects is considered. A HB approach using Markov chain Monte Carlo is used for the analysis of the U.S. state-level unemployment rate estimates for January 2004–December 2007. The sensitivity of such type of analysis to prior assumptions in the Gaussian context is also studied.

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1. Introduction

The unemployment rate is generally viewed as one of the key indicators of economic performance. The U.S. Bureau of the Labour Statistics (BLS) publishes monthly unemployment rate estimates for the national level as well as various geographic and demographic subdomains under Current Population Survey (CPS). The local unemployment rates are used in regional planning and fund allocation under various federal assistance programs.

The CPS uses a 4–8–4 rotating panel design which is conducted by the U.S. Census Bureau on a monthly basis. For any given month, the CPS sample can be grouped into eight subsamples corresponding to the eight rotating panels where all of the units belonging to a particular rotating panel enter and leave the sample at the same time. A given rotating panel remains in the sample for four consecutive months, stays out of the sample during the eight succeeding months, and then returns for another four succeeding months. It is then completely dropped from the sample and is replaced by a group of nearby households. For any given month, two new rotation groups are sampled, one is completely new and the other is a returning group, which has been out of the sample for 8 months. Hence in the CPS design, of the eight rotation groups, six are common between any two consecutive months (75% overlap) and four are common to the same month in consecutive years (50% overlap).

In the CPS design, many states do not have large enough samples to produce adequate direct estimates. To borrow strength from related states, (Datta et al., 1999), henceforth DLML, proposed a hierarchical Bayes (HB) method by using a hierarchical cross-sectional and time series model (see also Rao, 2003). However, this model does not account for spatial effects, geographical variations, and this factor is likely to be important. A comprehensive HB method to account for all the variations including spatial and temporal effects is studied.

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In Section 2, the DLML model is briefly reviewed. The model incorporating spatial effects is also described in detail. We discuss Bayesian inference using Markov chain Monte Carlo (MCMC) methods and techniques for model comparison in Section 3. A data analysis using the CPS time series data from January 2004 to December 2007 for U.S. states is presented in Section 4 with a corresponding evaluation of sensitivity of such type of analysis to prior assumptions in the Gaussian context. Concluding remarks are given in Section 5.

2. Statistical models

2.1. DLML model

Let y_{it} denote the CPS unemployment rate estimate for the i th state at time t . Define

$$y_{it} = \theta_{it} + e_{it}, \quad i = 1, \dots, I; \quad t = 1, \dots, T, \quad (2.1)$$

where θ_{it} is the true unemployment rate for the i th state at time t and e_{it} is the corresponding sampling error with mean 0 and variance σ_{it}^2 . Because the CPS design induces significant overlaps of the sampling units over time, it is important to account for a correlation between e_{it} and $e_{it'}$. We denote the corresponding covariance by $\sigma_{itt'}$ ($i = 1, \dots, I$; $t, t' = 1, \dots, T$; $t \neq t'$). Let $y_i = (y_{i1}, \dots, y_{iT})'$, $\theta_i = (\theta_{i1}, \dots, \theta_{iT})'$, and $e_i = (e_{i1}, \dots, e_{iT})'$, where e_i follows a multivariate Gaussian with mean vector 0 and sampling covariance matrix Σ_i . We then have

$$y_i \sim N(\theta_i, \Sigma_i), \quad i = 1, \dots, I.$$

The sampling covariance matrix Σ_i is unknown in the model, however, a smoothed estimator of the sampling variances is often used. There are methods to calculate the sampling covariances, for example generalized variance function (Tiller, 1989, 1992), equal coefficient of variation (CV) and equal design effects (You, 2008). Datta et al. (1999) used generalized variance function to produce sampling variances (diagonal elements of Σ_i). They used the sampling lag correlation and variance estimates to obtain the estimates of Σ_i . Note that (Datta et al., 1999) used 49 states and the District of Columbia for the analysis. They excluded the state of New York, since the unemployment insurance (UI) rate data, calculated as the ratio of the number of persons applying UI benefit over the number of persons in the labor force, used as auxiliary information, x_{it} , was very unreliable for the state of New York. Datta et al. (1999) developed the following model for θ_{it}

$$\theta_{it} = x_{it}\beta_i + v_i + \sum_{u=1}^{12} f_{itu}\gamma_{iu} + \sum_{w=1}^W g_{itw}\zeta_{iw} + \alpha_{it}, \quad (2.2)$$

where v_i and β_i are the state-specific intercept and slope, x_{it} is UI rate as auxiliary variable, and α_{it} is a random walk (RW) state-specific term. More precisely, they took $v_i \sim N(v, r_v^{-1})$ and $\beta_i \sim N(\beta, r_\beta^{-1})$, where r_v and r_β are the precision parameters of v_i and β_i . As the CPS estimates y_{it} are not seasonally adjusted, the seasonally adjusted terms $\sum_{u=1}^{12} f_{itu}\gamma_{iu} + \sum_{w=1}^W g_{itw}\zeta_{iw}$ used in the model. Let f_{itu} be the u th month indicator variable defined as $f_{itu} = 1$ if $t = u \bmod 12$ and $f_{itu} = 0$ otherwise; and $f_{it12} = 1$ if $t = 12, 24, \dots$ and $f_{it12} = 0$ otherwise. Similarly, let g_{itw} be the indicator variable for the w th year, defined as $g_{itw} = 1$ if $12(w-1) < t \leq 12w$ and $g_{itw} = 0$ otherwise. The usual restrictions, $\gamma_{i12} = -\sum_{u=1}^{11} \gamma_{iu}$ and $\zeta_{iW} = -\sum_{w=1}^{W-1} \zeta_{iw}$, were imposed to have a model with full rank; see Datta et al. (1999) for more details.

2.2. Spatio-temporal model

We develop a model for true unemployment rate θ_{it} with borrowing strength from different sources. Our model accommodates the spatially correlated and temporal random effects. The following model is considered for θ_{it}

$$\theta_{it} = m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \eta_i + \phi_i + \psi_{it}, \quad (2.3)$$

where m is a fixed effect representing the overall mean rate over time and state, x_{it} is an explanatory variable (e.g., UI) with the fixed parameter β_i , η_i and ϕ_i represent specified and unspecified features of state i that do and do not display spatial structure, respectively, and ψ_{it} is the interaction between space and time (Knorr-Held, 2000). The seasonal effects are introduced by $A_t \cos(\pi t/6) + B_t \sin(\pi t/6)$. The expressions $\cos(\pi t/6)$ and $\sin(\pi t/6)$ account for the seasonal variations over time with coefficients A_t and B_t which follow RW model (Fanshawe et al., 2008). This formulation allows the seasonal pattern to change over time, in contrast with DLML model where the components must sum exactly to 0. Alternatively, one may introduce the seasonal variation with applying the seasonal components $(s_1, \dots, s_{11})$ proposed by Knorr-Held and Richardson (2003) such that the components add up to Gaussian white noise over a moving window of length 12 months.

The first-order RW model is employed for the temporal effects A_t and B_t . The simple model RW(1) for the random effects A_t may be written as $A_t | A_{t-1} \sim N(A_{t-1}, \sigma_A^2)$, where σ_A^2 is an unknown scalar to be estimated from the data (Knorr-Held, 2000). The $(A_1, \dots, A_T)'$ is assumed to be multivariate Gaussian with mean vector 0 and precision matrix $\sigma_A^{-2} \mathbf{K}_T$, where $\mathbf{K}_T$ is a known structure matrix. Similar expressions are applied for B_t .

The usual intrinsic conditionally autoregressive (ICAR) model is used to capture the spatial random effects η_i . A variety of ICAR models may be used by taking a collection of mutually compatible conditional distributions $p(\eta_i | \eta_{-i})$, $i = 1, \dots, I$ where $\eta_{-i} = \{\eta_j : j \neq i, j \in \partial_i\}$ and ∂_i denote a set of neighbors for the i th state (Besag et al., 1991). We consider the following

general model for the spatial effects η_i ,

$$\boldsymbol{\eta} = (\eta_1, \dots, \eta_I)' \sim N(\mathbf{0}, \boldsymbol{\Sigma}_\eta), \quad (2.4)$$

$$\boldsymbol{\Sigma}_\eta = \sigma_\eta^2 \mathbf{D}^{-1},$$

where σ_η^2 is the spatial variance parameter. The neighborhood matrix $\mathbf{D}$ has its i th diagonal element equal to the number of neighbors of the corresponding state ($\# \partial_i$), and the off-diagonal elements in each row equal -1 if the corresponding states are neighbors and 0 otherwise (Leroux et al., 1999; MacNab and Dean, 2000, 2001). Neighbors can be defined in various ways, depending on the context of the analysis, but one popular definition is simply the set of states that have common borders and we use this definition.

The unstructured heterogeneity $\phi = (\phi_1, \dots, \phi_I)'$ is assumed to be Gaussian distribution with mean 0 and covariance matrix $\sigma_\phi^2 I_I$, where I_I is an identity matrix with dimension I .

The interaction effect of space and time ψ_{it} may be defined in many different ways. One way to define ψ_{it} is

$$\boldsymbol{\psi} = (\psi_{11}, \dots, \psi_{IT})' \sim N(0, \sigma_\psi^2 \mathbf{K}_\psi^{-1}),$$

where σ_ψ^2 measures the variance between space and time and $\mathbf{K}_\psi$ is a pre-specified structure matrix. The model (2.3) reduces to model with the main effects space and time when $\psi_{it} = 0$. Hence, ψ captures only the variation that is not explained by the main effects. Clayton (1996) suggested specifying $\mathbf{K}_\psi$ as the Kronecker product of the structure matrices of those main effects, which are assumed to interact.

3. Bayesian inference and model comparison

With advances in computational power, much progress in empirical Bayes and hierarchical Bayesian modeling has been made. It enables us to provide stable estimators for incidence rates in small areas by using information from all areas. A comprehensive account of hierarchical Bayesian models for spatial and spatio-temporal data is given by Banerjee et al. (2004).

The Bayesian approach is employed to estimate the model parameters $\omega = (\beta', \zeta')'$, where $\beta = (m, \beta_1, \dots, \beta_I)'$, $\zeta = (\sigma_A^2, \sigma_B^2, \sigma_\eta^2, \sigma_\phi^2, \sigma_\psi^2)'$. In the HB approach, we estimate ω by its posterior mean. The uncertainty in the HB estimate is measured by the posterior standard deviation of ω . The Gibbs sampler (e.g., Gelman and Rubin, 1992 and Gelfand and Smith, 1990) may be used to obtain the posterior mean and posterior variance of ω . The Gibbs sampler is a Monte Carlo Markovian updating scheme that gives the marginal, conditional, and joint distribution of the random variables $v_1, \dots, v_s$ (say), where $s = p + q$ with p and q as length of β and ζ , respectively. It is assumed that the joint density of $v_1, \dots, v_s$ is uniquely determined by the collection of conditional densities of v_r given $v_{-r} = (v_1, \dots, v_{r-1}, v_{r+1}, \dots, v_s)'$, $f(v_r | v_{-r})$, $r = 1, \dots, s$. The Gibbs sampler requires generation of samples from the conditional densities $f(v_r | v_{-r})$ for $r = 1, \dots, s$. Starting with an initial set of values $v_1^{(0)}, \dots, v_s^{(0)}$, a random number, $v_1^{(1)}$, is first generated from $f(v_1 | v_{-1}^{(0)})$. In a similar way, we generate $v_2^{(1)}, \dots, v_s^{(1)}$ successively from $f(v_2 | v_1^{(1)}, v_3^{(0)}, \dots, v_s^{(0)})$, $\dots$, $f(v_s | v_{-s}^{(1)})$. Hence, the first replicate $v_1^{(1)}, \dots, v_s^{(1)}$ is obtained. We continue replicating, say R times, where $(v_1^{(R)}, \dots, v_s^{(R)})$ converges to $(v_1, \dots, v_s)$ in distribution under some regularity conditions.

To generate samples from the posterior distribution using MCMC method via the Gibbs sampler, we should sample from the full conditional distributions. Note that in our application, all of these full conditional distributions are standard distributions that can be easily sampled. To implement our application in the HB setup, we use the WinBUGS software (Spiegelhalter et al., 2004). The ICAR is used for spatial random effects as defined in Section 2. We use the proper gamma priors for the precision of spatial effects, $(\sigma_\eta^{-2}, \sigma_\phi^{-2})$, where the proper gamma is given by $\sigma^{-2} \sim G(a, b)$ (say), with mean a/b and variance a/b^2 , to avoid problems with improper hyperpriors. Gamma priors are computationally convenient as the full conditional of σ^{-2} will again be gamma. The proper gamma prior is also considered for $\sigma_A^{-2}, \sigma_B^{-2}, \sigma_\psi^{-2}$. When considering $x_{it}\beta_i$ as a linear trend, the β_i 's would be considered as an independent Gaussian distribution with zero mean and variances σ_β^2 , where proper gamma priors are used for σ_β^{-2} .

In order to ensure that the ICAR model is identifiable, Besag and Kooperberg (1995) suggest constraining the random effects to sum to zero, and using a flat prior on the whole real line for the intercept m . For more details on the choice of hyperparameters in spatial modeling, see (Bernardinelli et al., 1995), the discussion in Best et al. (1999), and Eberly and Carlin (2000).

An important fact is to choose the best model among postulated sub-models of model (2.3) based on some criteria. Note that some criteria such as Bayes factor are not applicable to interpret the model in the presence of flat or ICAR priors. We have some measures of model comparison to be calculated easily with MCMC methods. In the context of Gaussian likelihood, the following three criteria will be compared in the application. First, deviance residual (McCullagh and Nelder, 1989) as

$$DR = \sum_{i=1}^I \sum_{t=1}^T \text{sgn}(y_{it} - \hat{\theta}_{it}) \sqrt{d_{it}},$$

where

$$d_{it} = (y_{it} - \hat{\theta}_{it})^2$$

and

$$\text{sgn}(z) = \begin{cases} 1 & z > 0 \\ 0 & z = 0 \\ -1 & z < 0 \end{cases}.$$

The second criterion is *saturated deviance* (McCullagh and Nelder, 1989) as

$$SD = \sum_{i=1}^I \sum_{t=1}^T d_{it}.$$

The last measure is *deviance information criterion* (DIC) which is a standard measure in the Bayesian set up (Spiegelhalter et al., 2002). This measure is a generalization of the Akaike information criterion, which handles hierarchical Bayesian models of any degree of complexity. It is given by

$$DIC = 2\overline{SD(\omega)} - SD(\overline{\omega}),$$

where $\overline{SD(\omega)}$ is posterior mean of saturated deviance and $\overline{\omega}$ is the posterior mean of model parameter vector ω . Although we rely mainly on this measure for assessing models in our application, the other measures are also computed for comparison. A smaller value of above criteria indicates a better fitting model.

4. Data analysis

4.1. Data description, implementation and estimation

A monthly unemployment CPS rate estimate of the U.S. states from January 2004 to December 2007 is considered. During the study period, the population of U.S. increased from 291.5 million in 2004 to 302.6 million in 2007. U.S. consists of 51 states, 50 states and the District of Columbia, and these states are the geographic unit used in our model and all data are linked to these geographic boundaries. The monthly average of the states sample size of labour forces ranges from 117.6 to 650.8, with mean and median values of 216.8 and 183.3, respectively. The largest sample belongs to California while Mississippi has the least sample. Also, the monthly average of direct estimate of unemployment rates per state ranges from 0.028 to 0.070 with the mean and median 0.047. The monthly average of CV value of the states ($\overline{CV}_i$) ranges from 6.10% to 21.35%, with mean and median values of 14.46% and 13.98%, respectively. The monthly average beneficiary rate of UI per state ranges from 0.001 to 0.413 with mean and median values of 0.022 and 0.015, respectively.

The UI beneficiary rates are used as auxiliary data, x_{it} , in the model. To obtain a smoothed estimate of the sampling covariance matrix Σ_i used in the model, we use the *equal design effects modeling approach*. To this end, we first obtain $\hat{\sigma}_{it}^2 = (\overline{CV}_i)^2 y_{it}^2$. The design effect, called *de*, for the i th state at time t is approximately given by

$$de_{it} \approx \frac{\sigma_{it}^2}{\theta_{it}(1-\theta_{it})/n_{it}},$$

where n_{it} is the corresponding sample size. Let $\hat{\gamma}_{it} = \hat{\sigma}_{it}^2/[y_{it}(1-y_{it})]$ be the estimate of $\gamma_{it} = de_{it}/n_{it}$. A smoothed average factor $\bar{\gamma}_i$ is obtained as $\bar{\gamma}_i = \sum_{t=1}^T \hat{\gamma}_{it}/T$. We then have a smoothed estimate $\hat{\Sigma}_i$ with diagonal elements $\hat{\sigma}_{it}^2 = \theta_{it}(1-\theta_{it})\bar{\gamma}_i$ and off-diagonal elements equal to $\hat{\sigma}_{itt'} = \bar{\rho}_{|t-t'|}(\sqrt{\hat{\sigma}_{it}^2 \hat{\sigma}_{it'}^2})$, where $\bar{\rho}_{|t-t'|}$ is the average correlation coefficient of lag $|t-t'|$ over time and states, and then treat $\hat{\Sigma}_i$ as the true Σ_i . We found that using the smoothed estimate of Σ_i in the model can significantly improve the estimate in terms of CV reduction.

Fig. 1 presents the crude national unemployment rate over period, which clearly shows the need of seasonal adjustment. There is also an overall decreasing trend in unemployment rate over period. We fit several spatio-temporal models of form (2.3) for analysing the unemployment rate estimates. For instance, the simple model (M1) is considered and defined as $\theta_{it} = m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6)$. Note that for model M1 and other models considered, we independently simulate $l=3$ parallel runs, each of length $H=2h$, with $h=7500$. To reduce the effects of the starting values on the final results, the first 7500 iterations of each run are deleted. We take every 5th iteration of the remaining 7500 iterations to reduce the autocorrelation in the run, leading to 1500 iterations for each run. Hence, we have $l=3$ runs with sample size $n=1500$ for each run. To monitor the convergence of the model parameters, we used several diagnostic methods implemented in the Bayesian output analysis (BOA) program (Smith, 2007), a freely available package created for software R (R Development Core Team). None of them showed any violation in terms of convergence of the model parameters.

We evaluate two different models for the interaction between space and time ψ_{it} ; in the first one, $(\psi_{1t}^{(1)}, \dots, \psi_{lt}^{(1)})'$ represents an ICAR model, and in the second one, $(\psi_{1t}^{(2)}, \dots, \psi_{lt}^{(2)})'$ represents a RW(1) model as suggested by Datta et al. (1999). We report the model M1 and other models in Table 1 which are found useful in our exploration of the data. These

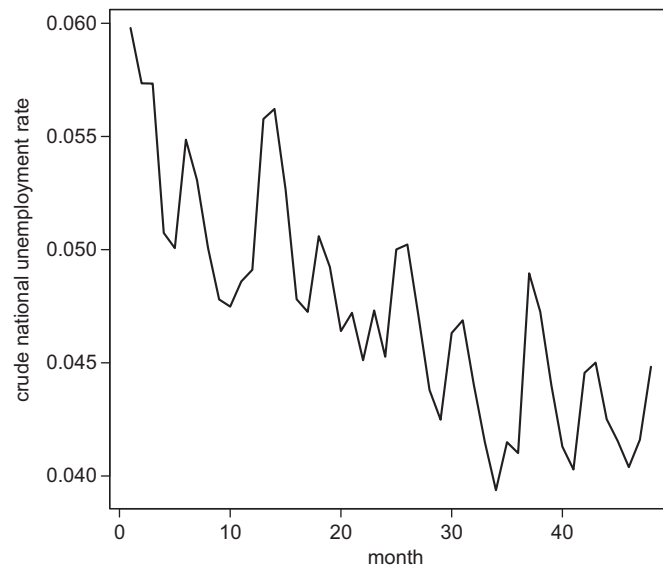


Fig. 1. Crude national unemployment rate over the period January 2004 to December 2007.

Table 1

The values of DR, SD and DIC for comparison of the sub-models of model (2.3).

Models defined from θ_{it}	DR	SD	DIC
M1 : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6)$	13.8	15.0	28.9
M2 : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \eta_i + \phi_i$	5.5	8.0	15.7
M2A : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \eta_i$	7.5	6.2	12.0
M2B : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \phi_i$	4.0	6.6	12.9
M3 : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \psi_{it}^{(1)} + \eta_i + \phi_i$	4.9	8.7	17.0
M3A : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \psi_{it}^{(1)} + \eta_i$	6.5	6.7	13.1
M3B : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \psi_{it}^{(1)} + \phi_i$	5.0	7.3	14.1
M4 : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \psi_{it}^{(2)} + \eta_i + \phi_i$	5.6	11.6	22.8
M4A : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \psi_{it}^{(2)} + \eta_i$	5.6	10.1	19.7
M4B : $m + x_{it}\beta_i + A_t \cos(\pi t/6) + B_t \sin(\pi t/6) + \psi_{it}^{(2)} + \phi_i$	4.7	10.7	21.0

sub-models of the spatio-temporal model (2.3) have an increasing level of complexity. For instance, model M2 is the simple model M1 plus spatially structured (η_i) and unstructured (ϕ_i) components. Model M3 is model M2 plus an interaction effect $\psi_{it}^{(1)}$. Model M4 is essentially model M3 with replacing $\psi_{it}^{(1)}$ by $\psi_{it}^{(2)}$. Sub-models of M2, M3 and M4 are variants of models M2, M3 and M4.

In terms of prior assumptions, we use a flat prior on the whole real line for m and highly dispersed priors for all other model parameters. In particular, we consider $N(0, \sigma_\beta^2)$ for the regression coefficients $\beta_i (i = 1, \dots, I)$ and $G(0.5, 0.0005)$ for the spatial hyperparameters $\sigma_\eta^{-2}, \sigma_\phi^{-2}$ and $\sigma_\psi^{-2}, \sigma_A^{-2}, \sigma_B^{-2}, \sigma_\beta^{-2}$.

To compare competing models, we calculate the measures of overall fit SD, DR and DIC. On the basis of these values listed in Table 1, model M2A is better than M2, particularly over M1 which includes no spatial random effects. There is evidence of spatial and temporal (seasonal) variations in unemployment data set. It seems that model M2A is better than models M3 and M4 in terms of the overall fit measures, noting that there is no striking evidence that one of these variants is substantially better than others. However, based on DIC criterion which is a standard measure in the Bayesian set up, the model M2A has the best performance compared to other models. We should also point out that comparing M3 and M4, the model M3 or one of its variants is better than M4, showing that the interaction effect $\psi_{it}^{(1)}$ has better fit than $\psi_{it}^{(2)} = \alpha_{it}$. Note that we also fitted the model (2.2) with priors as described in Datta et al. (1999), and resulted DR=9.2, SD=20.9 and DIC=41.0.

Fig. 2 gives the residuals, $dr_{it} = \text{sgn}(y_{it} - \hat{\theta}_{it})\sqrt{\hat{d}_{it}}$, versus predicted diagnostic plot for model M2A. It is clear from Fig. 2 that there is no serious lack of fit in our model. To have a comparison between model-based M2A and CPS design-based approach, we calculate the percent reductions of the standard errors of the HB estimates for model M2A over the CPS standard errors. The percent reduction is defined as the difference of the posterior standard deviation and the CPS standard error relative to the CPS standard error and is expressed as a percentage.

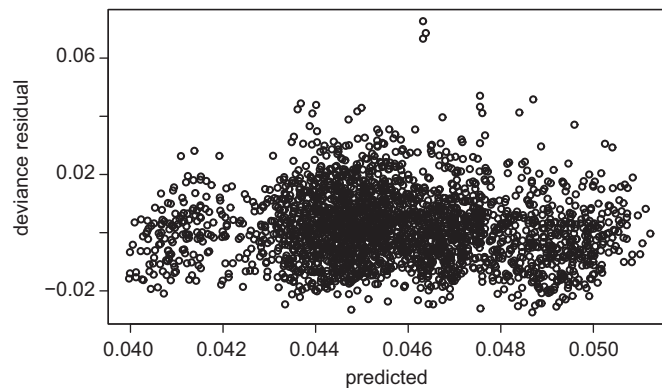


Fig. 2. The residuals versus predicted diagnostic plot of unemployment rate for model M2A over the period January 2004 to December 2007.

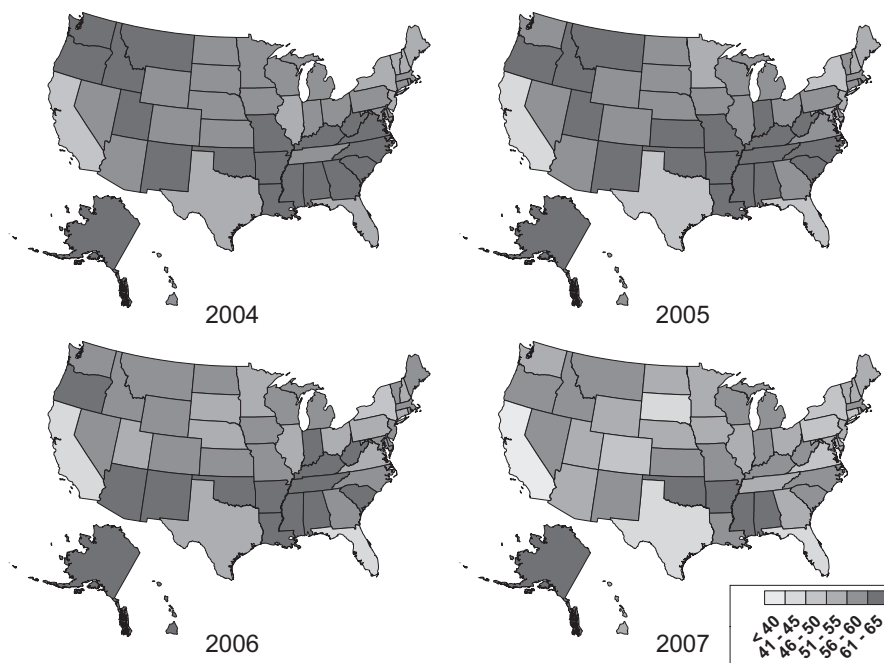


Fig. 3. Statewise percent reduction of standard errors for HB estimates of model M2A over the CPS standard errors of unemployment rates in June 2004–2007.

The *percent reduction measure* is to compare different model-based estimates and design-based CPS estimates, and we need to make some cautionary remarks. In general, the posterior variance under a given hierarchical model and the CPS variance estimate do not estimate the same measure of uncertainty and thus are not strictly comparable. The same comment applies regarding the comparison of posterior variances under different hierarchical models (Datta et al., 1999).

Fig. 3 shows the statewise percent reduction of the HB estimates over the CPS estimates for June (for example) 2004–2007. It is clear from these maps that the HB method is much better than the CPS for all states in terms of the variation of the estimates of unemployment rates. These maps suggest that the pattern of reduction of the HB estimates over the CPS estimates is relatively stable over time. There are also similar patterns for other months and years.

In Bayesian estimation, it is also important to choose suitable values for prior assumptions. Details of the prior sensitivity and choice of models are given in Pascutto et al. (2000). We investigate the choice of prior through a sensitivity study for the unemployment data set.

4.2. Prior sensitivity

In this section, the hyperprior distributions of the model parameters are evaluated. The hyperprior distributions of the variance components are generally set to be vague to allow learning from the data. Generally, the prior for the precision of

Table 2

The values of DR, SD and DIC for sensitivity analysis of prior distributions.

Prior	Models	DR	SD	DIC
A	M1	13.8	15.0	28.9
	M2	5.5	8.0	15.7
	M3	4.9	8.7	17.0
	M4	5.6	11.6	22.8
B	M1	13.4	15.0	28.9
	M2	5.2	8.0	15.7
	M3	4.6	8.7	17.0
	M4	5.3	11.6	22.8
C	M1	13.4	14.5	28.7
	M2	5.2	7.5	15.5
	M3	4.6	8.2	16.8
	M4	5.3	11.1	22.6
D	M1	13.1	14.4	28.8
	M2	4.9	7.4	15.7
	M3	4.3	8.1	17.0
	M4	5.0	11.0	22.8
E	M1	13.8	15.4	29.3
	M2	5.5	8.3	15.8
	M3	4.9	9.1	17.2
	M4	5.6	12.0	23.0
F	M1	13.4	15.2	29.1
	M2	5.2	8.1	15.6
	M3	4.5	8.9	16.9
	M4	5.2	11.8	22.7
G	M1	13.6	14.6	29.0
	M2	5.3	7.5	15.7
	M3	4.6	8.2	17.0
	M4	5.3	11.1	22.8

the random effects (σ^{-2}) is set to be gamma distribution with scale and shape parameters equal to 0.001. However, to model the precision of the spatial random effects in a ICAR model, it is common to use a gamma distribution with scale parameter 0.5 and shape parameter 0.0005 (Bernardinelli et al., 1995; Kelsall and Wakefield, 1999; Silva et al., 2008).

To investigate the influence of hyperprior specifications for the Gaussian context, we use a sensitivity analysis with respect to the prior distributions for the spatial precision parameters σ_{η}^{-2} and σ_{ϕ}^{-2} , assuming a variety of different gamma priors $G(a,b)$. The same priors are used for $\sigma_{\beta}^{-2}, \sigma_A^{-2}, \sigma_B^{-2}$ and σ_{ψ}^{-2} . We use the following combinations in our experimental design: $(a,b) = (0.5, 0.0005), (0.001, 0.001), (0.01, 0.01), (0.1, 0.1), (2, 0.001), (0.2, 0.0004)$ and $(10, 0.25)$, which are denoted by A, B, C, D, E, F and G, respectively. The prior B has smaller variance than prior A, and priors C and D are variants of prior B, with associated variance in decreasing order. Priors E and F correspond to distributions with the similar variance as prior A. The variance of prior G is between the variance of priors C and B.

We study the sensitivity of model selection with respect to the hyperparameters. The measures DR, SD and DIC are reported with respect to assumed priors. Table 2 provides summary measures of model fit for models M1, M2, M3 and M4 with various hyperparameter settings in the analysis of unemployment data set. Comparing DIC values across both models and priors, it seems that the more striking effect is that of the model fitted in the sense that DIC tends to prefer model M2 compared to other models, and there is a smaller difference in DIC across priors than across models. It seems that for vague priors considered here, the omitting main effects in the model play a more important role than the choice of prior in determining goodness of fit with respect to DIC and these other measures of fit considered. We also consider different combination of priors (A–G) for models M1, M2, M3 and M4 and the results are robust in terms of choosing priors (not shown here). Consequently, changing prior assumptions on the variance components does not have a considerable effect on the selected model in Section 4.1.

5. Concluding remarks

Datta et al. (1999) proposed a hierarchical Bayes model to obtain estimates of monthly unemployment rate of the U.S. states from January 1985 to December 1988. They applied the state-specific intercept and slope v_i and β_i , and random walk for temporal effects α_{it} . They used the indicator functions to capture the seasonal effects.

However, the geographical variation is also likely to be important. In this paper, we studied a hierarchical Bayes model which accounts for possible geographical and temporal variations. This model was used to obtain model-based estimates of monthly unemployment rate for the U.S. state level. The model borrows strength across states and over time periods simultaneously. The well-known intrinsic conditionally autoregressive model was employed for spatially correlated random effects. To account the seasonal effects, the functions $\cos(\cdot)$ and $\sin(\cdot)$ were used, which allow the seasonal pattern to change over time, in contrast with the indicator functions used in DLML model. The analysis showed that incorporating

intrinsic conditionally autoregressive approach into the model, to capture the geographical variations over states, fits the data quite well. The hierarchical Bayes estimates improved the direct estimates significantly in terms of *percent reduction measure*, especially for states with small sample sizes. We used the common CV approach to estimate the sampling covariance matrix Σ_i and then treated it as true Σ_i . The uncertainty associated with the estimation of Σ_i was ignored as we assumed that the sampling covariance matrix Σ_i is known in the model.

In this paper, we used a linear mixed linking model (2.3) for the parameters θ_{it} , which matches with the sampling model (2.1). The non-linear mixed model has been also used for linking model (You and Rao, 2002; You, 2008). We plan to study a log-linear mixed linking model for θ_{it} .

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References

- Banerjee, S., Carlin, B.P., Gelfand, A.E., 2004. Hierarchical Modeling and Analysis for Spatial Data. Chapman and Hall, CRC, London, Boca Raton, FL.
- Bernardinelli, L., Clayton, D., Montomoli, C., 1995. Bayesian estimates of disease maps: how important are priors? *Statist. Med.* 14, 2411–2431.
- Besag, J., Kooperberg, C.L., 1995. On conditional and intrinsic autoregressions. *Biometrika* 82, 733–746.
- Besag, J., York, J., Mollié, A., 1991. Bayesian image restoration, with two applications in spatial statistics. *Ann. Inst. Statist. Math.* 43, 1–21.
- Best, N.G., Arnold, R.A., Thomas, A., Waller, L.A., Conlon, E.M., 1999. Bayesian models for spatially correlated disease and exposure data (with discussion). In: Bernardo, J.M., Berger, A.P., Smith, A.F. (Eds.), *Bayesian Statistics*, vol. 6. Clarendon Press, Oxford, pp. 131–156.
- Clayton, D.G., 1996. Generalized linear mixed models. In: Gilks, W.R., Richardson, S., Spiegelhalter, D.J. (Eds.), *Markov Chain Monte Carlo in Practice*. Chapman and Hall, London, UK, pp. 275–301.
- Datta, G.S., Lahiri, P., Maiti, T., Lu, K.L., 1999. Hierarchical Bayes estimation of unemployment rates for the states of the U.S. *J. Amer. Statist. Assoc.* 94, 1074–1082.
- Eberly, L.E., Carlin, B.P., 2000. Identifiability and convergence issues for Markov chain Monte Carlo fitting of spatial models. *Statist. Med.* 19, 2279–2294.
- Fanshawe, T.R., Diggle, P.J., Rushton, S., Lurz, P.W.W., Glinianaia, S.V., Pearce, M.S., Parker, L., Charlton, M., Pless-Mulloli, T., 2008. Modelling spatio-temporal variation in exposure to particulate matter: a two-stage approach. *Environmetrics* 19, 549–566.
- Gelfand, A.E., Smith, A.F.M., 1990. Sampling-based approaches to calculating marginal densities. *J. Amer. Statist. Assoc.* 85, 398–409.
- Gelman, A., Rubin, D.B., 1992. Inference from iterative simulation using multiple sequences (with discussion). *Statist. Sci.* 7, 457–511.
- Kelsall, J.E., Wakefield, J.C., 1999. Discussion of Bayesian models for spatially correlated disease and exposure data, by Best N.G., Waller L.A., Thomas A., Conlon E.M., Stern R. In: Bernardo, J.M., Berger, J.O., Dawid, A.P., Smith, A.F.M. (Eds.), *Bayesian Statistics*, vol. 6. Oxford University Press, Oxford.
- Knorr-Held, L., 2000. Bayesian modeling of inseparable space-time variation in disease risk. *Statist. Med.* 19, 2555–2567.
- Knorr-Held, L., Richardson, S., 2003. A hierarchical model for space-time surveillance data on meningococcal disease incidence. *Appl. Statist.* 52, 159–183.
- Leroux, B.G., Lei, X., Breslow, N., 1999. Estimation of disease rates in small areas: a new mixed model for spatial dependence. In: Halloran, M.E., Berry, D. (Eds.), *Statistical Models in Epidemiology, the Environment and Clinical Trials*. Springer-Verlag, New York, pp. 135–178.
- MacNab, Y.C., Dean, C.B., 2000. Parametric bootstrap and penalized quasi-likelihood inference in conditional autoregressive models. *Statist. Med.* 19, 2421–2435.
- MacNab, Y.C., Dean, C.B., 2001. Autoregressive spatial smoothing and temporal spline smoothing for mapping rates. *Biometrics* 57, 949–956.
- McCullagh, P., Nelder, J.A., 1989. *Generalized Linear Models*, second ed. Chapman and Hall, London.
- Pascutto, C., Wakefield, J.C., Best, N.G., Richardson, S., Bernardinelli, L., Staines, A., Ellist, P., 2000. Statistical issues in the analysis of disease mapping data. *Statist. Med.* 19, 2493–2519.
- R Development Core Team: A Language and Environment for Statistical Computing, R Foundation for Statistical Computing. <<http://www.Rproject.org>>.
- Rao, J.N.K., 2003. *Small Area Estimation*. John Wiley, New York.
- Silva, G.L., Dean, C.B., Niyonsenga, T., Vanasse, A., 2008. Hierarchical Bayesian spatiotemporal analysis of revascularization odds using smoothing splines. *Statist. Med.* 27, 2381–2401.
- Smith, B.J., 2007. BOA: an R package for MCMC output convergence assessment and posterior inference. *J. Statist. Software* 21 (11), 1–37.
- Spiegelhalter, D.J., Best, N.G., Carlin, B.P., Van der Linde, A., 2002. Bayesian measures of model complexity and fit. *J. Roy. Statist. Soc. B* 64, 583–639.
- Spiegelhalter, D.J., Thomas, A., Best, N.G., Lunn, D.J., 2004. *WinBUGS version 1.4 User Manual*. MRC Biostatistics Unit, Institute of Public Health, London.
- Tiller, R., 1989. A kalman filter approach to labour force estimation using survey data. In: *Proceedings of the Survey Research Section, American Statistical Association*, pp. 16–25.
- Tiller, R., 1992. Time series modeling of sample survey data from the U.S. current population survey. *J. Official Statist.* 8, 149–166.
- You, Y., 2008. An integrated modelling approach to unemployment rate estimation for sub-provincial areas of Canada. *Survey Methodol.* 34, 19–27.
- You, Y., Rao, J.N.K., 2002. Small area estimation using unmatched sampling and linking models. *Can. J. Statist.* 30, 3–15.